

FATEME FOULADI

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AI/Machine Learning professional holding an M.S. in Computer Science (Artificial Intelligence) from the University of Tehran, with hands-on experience building LLM and machine learning applications. Experienced in RAG, conversational AI, prompt engineering, and AI backend development using Python, FastAPI, Gemini/OpenAI APIs, PostgreSQL, Docker, LangChain, embeddings, and vector search. Strong background in NLP, deep learning, computer vision, and graph learning.

AI & MACHINE LEARNING PROJECTS

AI Resume & Job Match Analyzer | [Live Demo](#) | [GitHub](#)

Python | FastAPI | Gemini API | PostgreSQL | SQLAlchemy | Alembic | Docker | Pydantic | Pytest | React | Vite | Railway | Netlify

Built and deployed an end-to-end, cross-industry AI resume–job matching system supporting text and PDF/DOCX resumes, with structured match scores, matched/partial/missing requirements, evidence, and targeted suggestions.

- Designed a multi-stage LLM pipeline for requirement extraction and evidence-based matching using schema-constrained outputs, prompt versioning, and primary/fallback Gemini models.
- Implemented deterministic evidence validation and weighted scoring to reduce hallucinated matches and make results more reproducible and explainable.
- Built an AI evaluation and observability framework tracking extraction precision/recall, match accuracy, grounding, latency, model usage, fallback recovery, and per-stage failures.
- Added PostgreSQL persistence, Alembic migrations, automated tests, Dockerized services, a React/Vite frontend, and cloud deployment with Railway and Netlify.

Conversational AI OrderBot for Restaurant Ordering

Python | OpenAI API | Panel | Prompt Engineering

- Developed a multi-turn restaurant ordering chatbot with structured outputs for order handling, delivery/pickup selection, and workflow automation.

RETRIEVAL-AUGMENTED QUESTION ANSWERING SYSTEM (RAG)

Python | LangChain | ChatOpenAI | DocArray | Vector Search

- Built a document QA system with semantic search and vector retrieval over structured product data to generate context-grounded answers.

MOLECULAR PROPERTY PREDICTION USING GRAPH NEURAL NETWORKS

PyTorch | GNNs | Deep Learning

- Developed and optimized graph-based molecular representation models, including a novel GNN architecture that improved over baseline methods.

CORONARY ARTERY STENOSIS SEGMENTATION IN X-RAY ANGIOGRAPHY

YOLO | Transformer | Mamba | PyTorch | Medical Image Analysis

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- Developed segmentation models for coronary artery stenosis in angiography images and contributed to a published IEEE paper using Mamba-based architectures.

Publication: "[Segmentation of Coronary Artery Stenosis in X-ray Angiography using Mamba Models](#)"

Authors: Fatemeh Fouladi, Ali Rostami, Hedieh Sajedi

EXPERIENCE

Machine Learning Bootcamp

Open AI School – University of Tehran | May 2023 - June 2023

- Built an oil-price forecasting model, a ResNet product image classifier, and a ParsBERT sentiment analysis system for Persian customer comments.
 - Performed data preprocessing, model training, evaluation, and result analysis using Python machine learning libraries.
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SKILLS

- Programming & Backend: Python, SQL, Bash, FastAPI, REST APIs, Pydantic, SQLAlchemy
 - Generative AI & NLP: Gemini API, OpenAI API, LLMs, RAG, Prompt Engineering, LangChain, LangGraph, Embeddings, Vector Search, Transformers, GPT, BERT, ParsBERT
 - Databases & DevOps: PostgreSQL, Alembic, Docker, Docker Compose, Git, Linux, SSH, Pytest
 - Machine Learning/ Deep Learning / Vision: PyTorch, TensorFlow, Scikit-learn, Regression, Classification, CNNs, ResNet, YOLO, U-Net, OpenCV, Medical Image Segmentation
 - Graph & Data: GNNs, Representation Learning, Pandas, NumPy, Matplotlib, Data Structures & Algorithms, Linear Algebra, Probability, Statistics
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EDUCATION

Master of Science in Computer Science – Artificial Intelligence

University of Tehran | 2023 – 2026

Relevant Coursework: Machine Learning, Natural Language Processing, Advanced Algorithms, Image Processing

Bachelor of Science (B.S.) in Mathematics

K. N. Toosi University of Technology | 2018 – 2022

COURSES AND CERTIFICATIONS

- **LangChain- Agentic AI Engineering with LangChain & LangGraph** | Udemy
 - **Machine Learning with Graphs** | Stanford University
 - **Natural Language Processing Specialization** | Coursera
 - **Yann LeCun's Deep Learning Course** | New York University
 - **Agentic AI courses** | DeepLearning.ai
 - **Machine Learning Bootcamp** | Open AI School
 - **Python for Data Science and Machine Learning Bootcamp** | Udemy
 - **Machine Learning by Stanford University** | Coursera
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